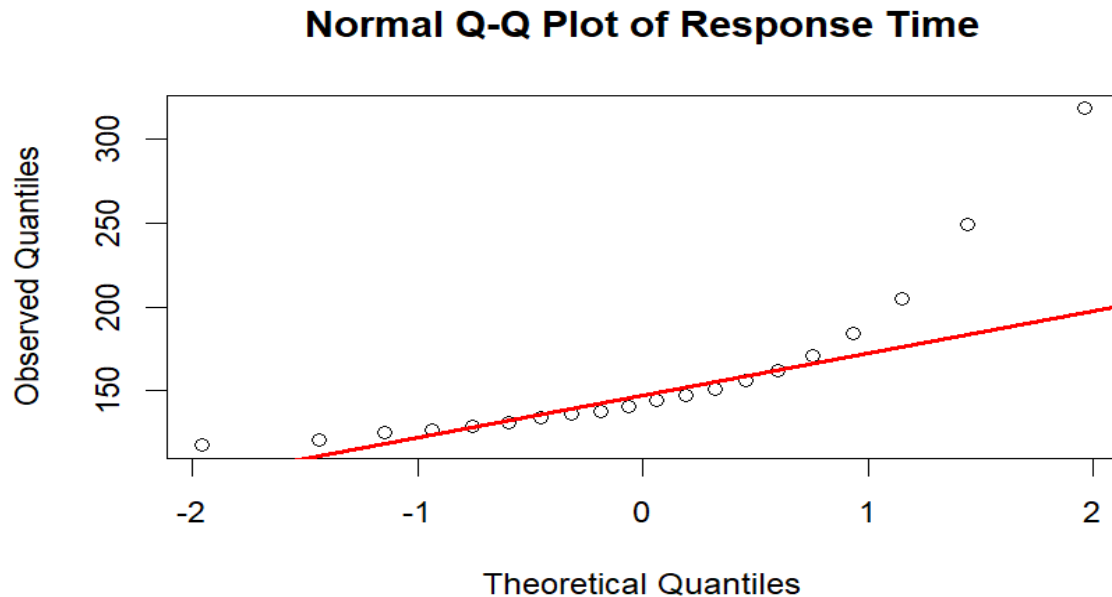


CO3_AT1_Assessment Tool: EXPERIMENT

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each experiment, use an appropriate visualization, justify the design, interpret the result, and identify potential visualization bias. R/ggplot2 or an equivalent visualization tool may be used.

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot



CODE:

```
# Dataset
response_time <- c(118,121,125,127,129,131,134,136,138,141,
                  144,147,151,156,162,171,184,205,249,318)
# 1. ECDF
plot(ecdf(response_time),
     main = "ECDF of Response Time",
     xlab = "Response Time (ms)",
     ylab = "Cumulative Probability",
     verticals = TRUE,
     do.points = TRUE)
```

```
# 2. Normal Q-Q Plot
qqnorm(response_time,
      main = "Normal Q-Q Plot of Response Time",
      xlab = "Theoretical Quantiles",
      ylab = "Observed Quantiles")
qqline(response_time, col = "red", lwd = 2)
# Basic statistics
mean(response_time)
sd(response_time)
median(response_time)
# Skewness
skewness <- mean((response_time - mean(response_time))^3) /
  (mean((response_time - mean(response_time))^2))^(3/2)
Skewness
```

Interpretation

- The ECDF shows that most response times are concentrated between approximately **118 ms and 184 ms**.
- The observations **205, 249 and 318 ms** form a long upper tail.
- The Q-Q plot will show deviations from the straight reference line, especially at the upper end.
- Therefore, the data **does not appear normally distributed**.
- The distribution has **positive/right skewness**.
- The skewness coefficient is approximately **2.08**, which indicates substantial right skewness.
- The value **318 ms** is an extreme observation.

Important statistics

Statistic	Value
Mean	159.35 ms
Median	142.50 ms
Standard deviation	49.00 ms
Skewness	≈ 2.08

Since the **mean is much greater than the median**, this also supports right skewness.

Recommendation

Percentile-based limits are preferable.

For example:

```
quantile(response_time, c(0.05, 0.95))
```

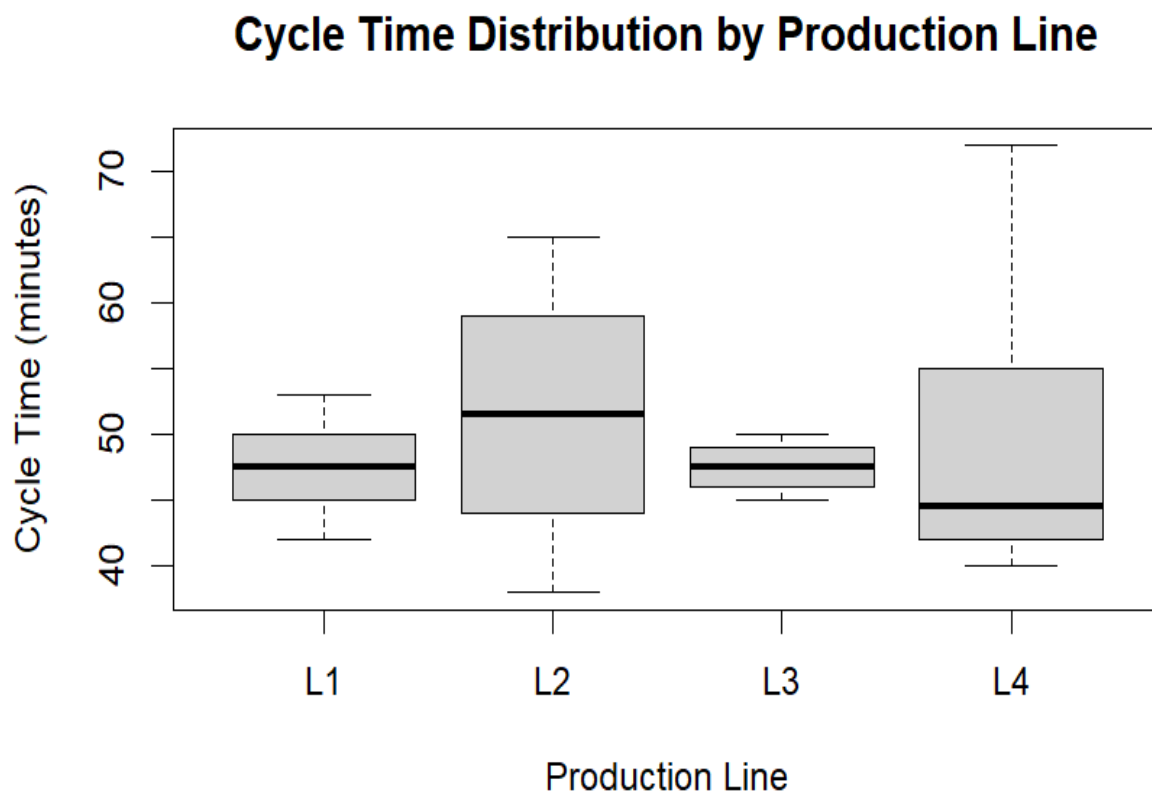
The approximate 5th and 95th percentiles are:

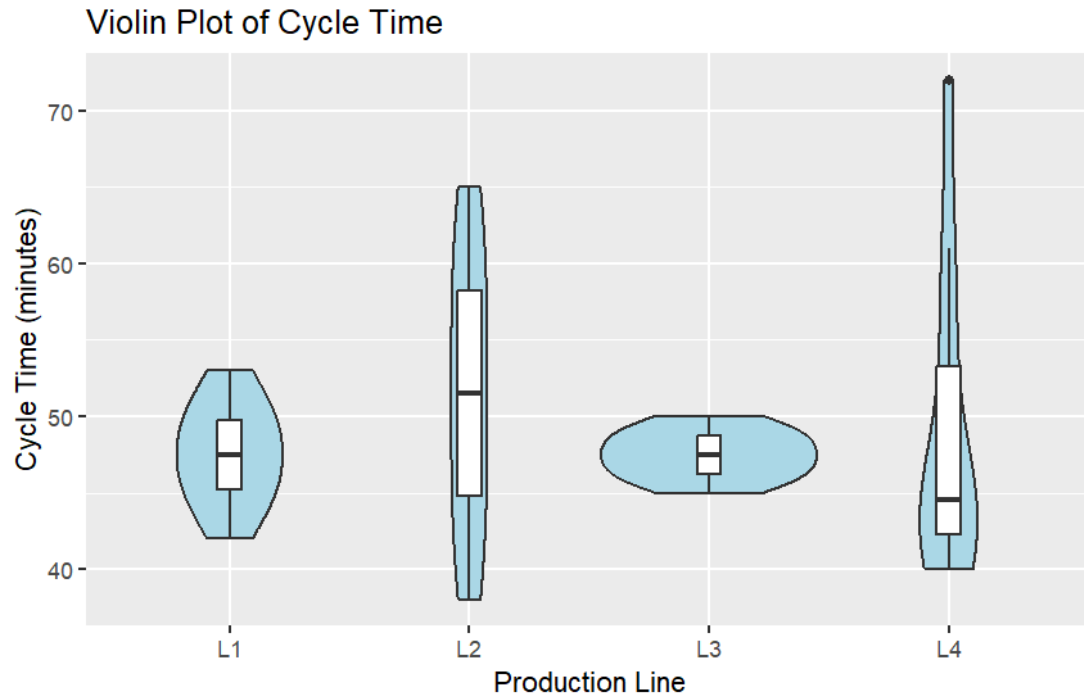
- 5th percentile $\approx$ **120.85 ms**
- 95th percentile $\approx$ **252.45 ms**

Using mean $\pm$ standard deviation would be less reliable because the extreme high values inflate both the mean and standard deviation.

Conclusion: Use percentile-based monitoring limits rather than mean $\pm$ standard deviation.

Question 2 – Comparing Many Distributions and Bean Plots





CODE:

```
# Data
L1 <- c(42,44,45,46,47,48,49,50,51,53)
L2 <- c(38,41,44,47,50,53,56,59,62,65)
L3 <- c(45,46,46,47,47,48,48,49,49,50)
L4 <- c(40,41,42,43,44,45,48,55,61,72)
cycle_time <- data.frame(
  Line = rep(c("L1","L2","L3","L4"), each = 10),
  Time = c(L1,L2,L3,L4)
)
# Boxplot comparison
boxplot(Time ~ Line,
  data = cycle_time,
  main = "Cycle Time Distribution by Production Line",
  xlab = "Production Line",
  ylab = "Cycle Time (minutes)")
# Bean-plot equivalent: violin plot
install.packages("ggplot2")
library(ggplot2)
ggplot(cycle_time, aes(x = Line, y = Time)) +
  geom_violin(fill = "lightblue") +
  geom_boxplot(width = 0.1) +
  labs(title = "Violin Plot of Cycle Time",
    x = "Production Line",
    y = "Cycle Time (minutes)")
```

Results

Line	Mean	SD	Range
L1	47.5	3.37	11
L2	51.5	9.08	27
L3	47.5	1.58	5
L4	49.1	10.44	32

Interpretation

Greatest variability: L4

- Highest standard deviation = 10.44
- Largest range = 32 minutes
- High values such as 61 and 72 indicate a long upper tail.

Most concentrated: L3

- SD = 1.58
- Range = only 5 minutes
- Values are tightly clustered around 48 minutes.

Why comparing only means is insufficient

Consider L1 and L3:

- L1 mean = 47.5
- L3 mean = 47.5

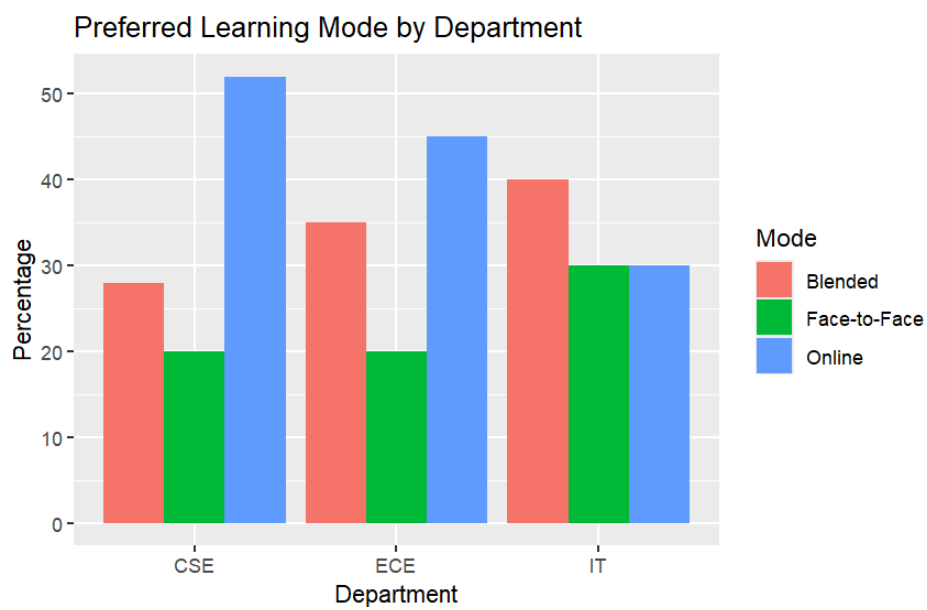
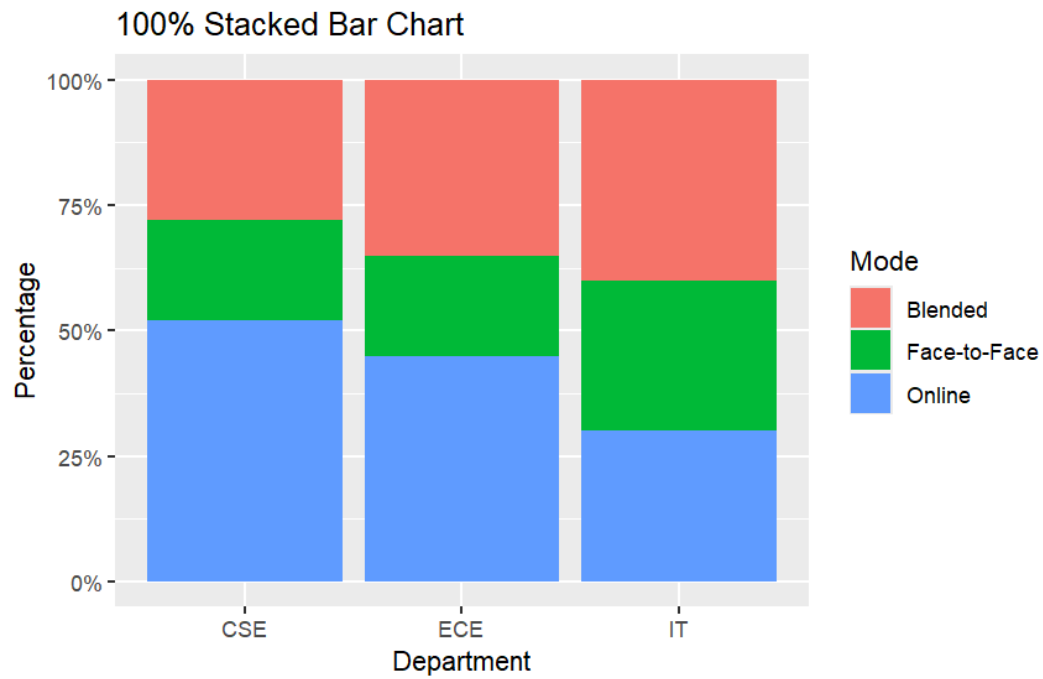
Their means are exactly the same, but their variability is very different.

Therefore, the mean alone cannot show:

- spread,
- variability,
- skewness,
- outliers,
- concentration of observations.

Conclusion: Distribution visualizations such as violin/bean plots and boxplots provide much more information than means alone.

Question 3 – Proportion Visualization and Choice of Chart



CODE:

```
# Data
```

```
learning <- data.frame(
```

```
  Department = c("CSE", "ECE", "IT"),
```

```

Online = c(52,45,30),

Blended = c(28,35,40),

Face_to_Face = c(20,20,30))

# Convert to long format

library(ggplot2)

learning_long <- data.frame(

  Department = rep(learning$Department, each = 3),

  Mode = rep(c("Online", "Blended", "Face-to-Face"), 3),

  Percentage = c(52,28,20,

                 45,35,20,

                 30,40,30))

# 1. Side-by-side bar chart

ggplot(learning_long,

       aes(x = Department, y = Percentage, fill = Mode)) +

  geom_bar(stat = "identity",

          position = "dodge") +

  labs(title = "Preferred Learning Mode by Department",

       x = "Department",

       y = "Percentage")

# 2. 100% stacked bar chart

ggplot(learning_long,

       aes(x = Department, y = Percentage, fill = Mode)) +

  geom_bar(stat = "identity",

          position = "fill") +

```

```
scale_y_continuous(labels = function(x) paste0(x*100, "%")) +  
labs(title = "100% Stacked Bar Chart",  
      x = "Department",  
      y = "Percentage")
```

Which chart is better?

For comparing **individual proportions**, the **side-by-side bar chart is better**.

For example, Online learning:

- CSE = 52%
- ECE = 45%
- IT = 30%

The bars have a common baseline, so these differences are easy to compare.

The 100% stacked chart is better for understanding the **overall composition within each department**, but individual middle segments are harder to compare because they do not share the same baseline.

Why separate pie charts are less effective

Separate pie charts require the viewer to compare angles or areas between different circles.

For example, comparing:

- CSE Online = 52%
- ECE Online = 45%
- IT Online = 30%

is more difficult using three separate pies than using aligned bars.

Misleading design choice

A designer could use a **truncated y-axis**, for example:

Y-axis: 40% to 55%

instead of starting at 0%.

This would make a relatively small difference such as **52% vs 45%** appear much larger.

Other misleading choices include:

- 3D effects
- unequal bar widths
- inappropriate scaling
- exaggerated colors
- distorted pie sizes

Question 4 – Nested Proportions using Treemap and Sunburst

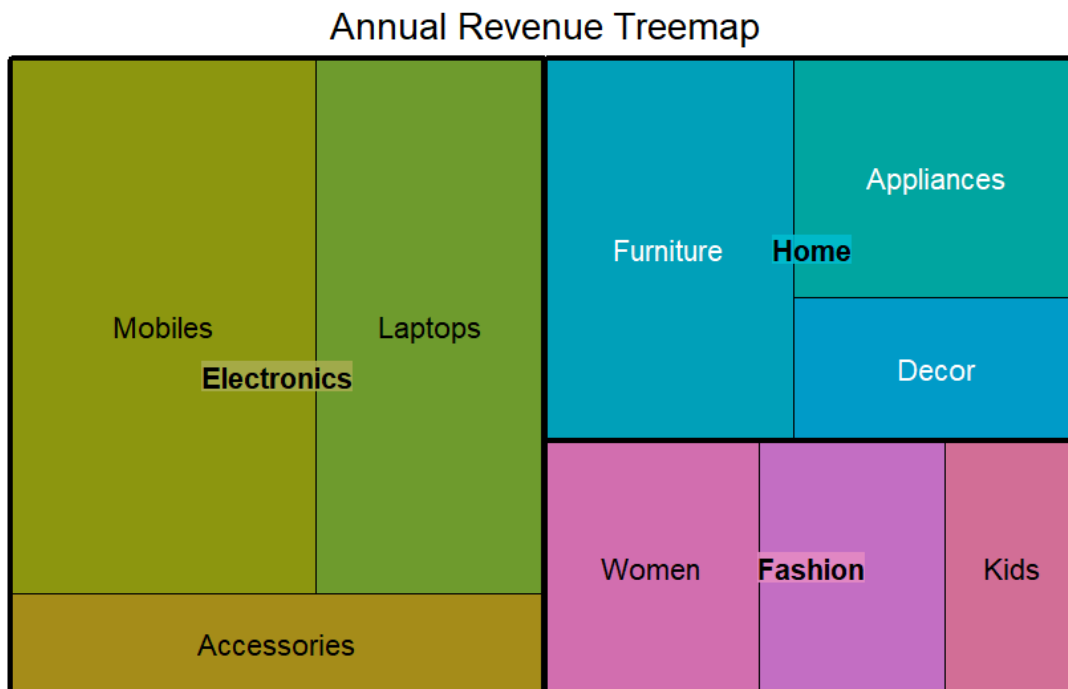
Important data issue

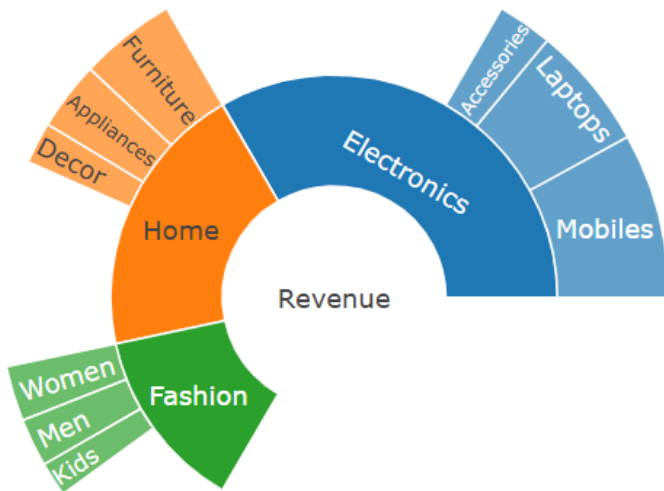
The provided data adds up to:

$$24 + 18 + 8 + 14 + 10 + 6 + 7 + 8 + 5 = 100 \text{ lakh}$$

So the **table contains ₹100 lakh = ₹1 crore**, not ₹1.2 crore.

Therefore, calculations based on the **provided data** should use ₹100 lakh as the total.





CODE:

```
install.packages("plotly")

library(plotly)

sunburst_data <- data.frame(

  labels = c(

    "Revenue",

    "Electronics", "Mobiles", "Laptops", "Accessories",

    "Home", "Furniture", "Appliances", "Decor",

    "Fashion", "Men", "Women", "Kids"

  ),

  parents = c(

    "",

    "Revenue", "Electronics", "Electronics", "Electronics",

    "Revenue", "Home", "Home", "Home",

    "Revenue", "Fashion", "Fashion", "Fashion"

  ),

)
```

```

values = c(
  100,
  50,24,18,8,
  30,14,10,6,
  20,7,8,5 ))

plot_ly(
  data = sunburst_data,
  labels = ~labels,
  parents = ~parents,
  values = ~values,
  type = "sunburst"
)

```

Category totals

Category	Revenue	Percentage
Electronics	₹50 lakh	50%
Home	₹30 lakh	30%
Fashion	₹20 lakh	20%
Total	₹100 lakh	100%

Largest category

Electronics is the largest category.

Its revenue:

$$24+18+8=50 \text{ lakh}$$

Percentage:

$$100/50 \times 100 = 50\%$$

So, based on the supplied table, **Electronics contributes 50% of revenue.**

If the stated ₹1.2 crore total is intended to be correct, the dataset is incomplete because the listed subcategories account for only ₹1 crore. In that case, Electronics would be $50/120 \times 100 = 41.67\%$ of the stated total.

Treemap vs Sunburst

Treemap	Sunburst
Uses rectangular areas	Uses concentric rings
Good for comparing overall sizes	Good for showing hierarchy
Uses space efficiently	Clearly shows parent-child relationships
Large categories are easy to identify	Nested composition is visually intuitive

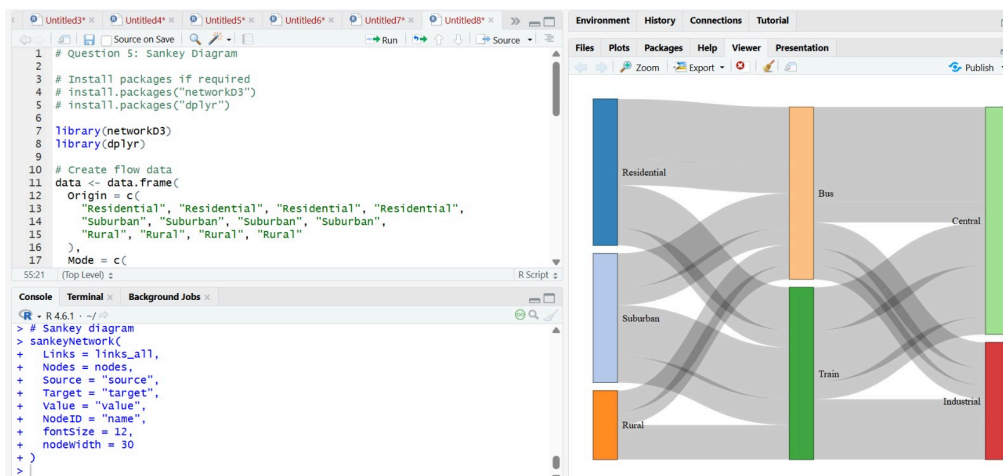
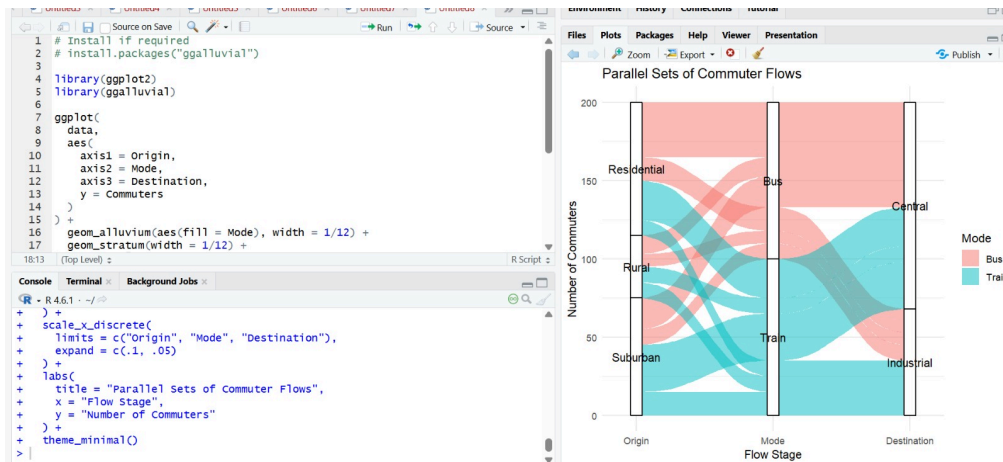
When can area encoding mislead?

Area comparisons can be difficult because humans are generally better at comparing **lengths** than areas.

For example, two categories with areas of 20 and 25 units may not visually appear as different as their actual values.

Very small rectangles can also become difficult to distinguish.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets



CODE:

```
install.packages("networkD3")
library(networkD3)
# Links
links <- data.frame(
  source = c(
    0,0,1,1,2,2,3,3,4,4,5,5,
    0,0,1,1,2,2
  ),
  target = c(
    3,4,3,4,3,4,3,4,3,4,3,4,
    6,7,6,7,6,7
  ),
  value = c(
    35,15,25,10,20,10,30,15,12,8,10,10,
```

```

    0,0,0,0,0,0
  )
)

library(networkD3)
nodes <- data.frame(
  name = c(
    "Residential", "Suburban", "Rural",
    "Bus", "Train",
    "Central", "Industrial"))

links <- data.frame(
  source = c(0,0,1,1,2,2,3,3,4,4),
  target = c(3,4,3,4,3,4,5,6,5,6),
  value = c(50,35,30,45,20,20,122,58,65,35))
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 30
)Three largest flows

```

The individual flows are:

Flow	Commuters	Percentage
Residential → Bus → Central	35	17.5%
Suburban → Train → Central	30	15%
Residential → Train → Central	25	12.5%

Calculation:

$20035 \times 100 = 17.5\%$ $20030 \times 100 = 15\%$ $20025 \times 100 = 12.5\%$

Together:

$17.5 + 15 + 12.5 = 45\%$

So the three largest individual flows represent **45% of all commuters**.

Sankey vs Parallel Sets

Sankey Diagram	Parallel Sets
Excellent for showing flow magnitude	Excellent for categorical relationships
Link width represents quantity	Band width represents quantity
Easy to identify dominant flows	Easy to follow categories across dimensions
Good for bottleneck identification	Good for comparing multiple categorical dimensions
Can become cluttered with many flows	Can also become crowded with many categories

Misleading design choice

If the link widths in a Sankey diagram are **not proportional to the actual commuter counts**, the viewer may incorrectly perceive a smaller flow as dominant.

Similarly, poor category ordering in a parallel-sets chart can cause lines to cross unnecessarily, making relationships difficult to follow.

Question	Main Finding
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Q1 Response time is right-skewed, not approximately normal; use percentile limits

Q2 L4 has greatest variability; L3 is most concentrated

Q3 Side-by-side bars are best for comparing individual proportions

Q4 Electronics = ₹50 lakh = 50% of the supplied ₹100 lakh data

Q5 Largest flow is Residential → Bus → Central = 35 = 17.5%

Important points

1. **ECDF** shows the cumulative proportion of observations at or below each value.
2. **Q-Q plot** compares observed quantiles with theoretical normal quantiles.
3. A straight Q-Q pattern supports normality; systematic deviations indicate non-normality.
4. **Right skewness** means the distribution has a longer tail toward larger values.
5. **Violin/bean plots** show distribution shape as well as spread.
6. **Side-by-side bars** are better for comparing individual percentages.
7. **Treemaps and sunbursts** are useful for hierarchical/nested proportions.
8. **Sankey diagrams** use link width to represent flow magnitude.
9. **Mean alone is insufficient** because two datasets can have the same mean but very different variability.
10. Always check whether the **visual encoding accurately represents the underlying data**.